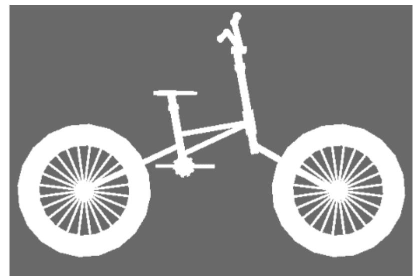
DATA SCIENCE 2VG3 Homework 1 2025/2026
Due: End of Day, Friday, 30th January 2026, via Avenue

Panda3d Exercise Bike

You will generate python scripts that perform the requested exercises and submit them via avenue. You should submit your best effort that actually runs rather than one program for each exercise (i.e. Exercise 5 only). If you submit multiple files, we will only look at the last submission.

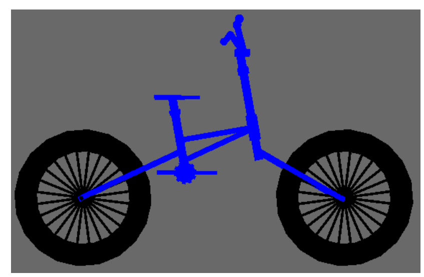
Exercise 1 Basic Scene



I have provided two model components, a wheel (wheel.egg) and a bicycle frame (frame.egg). You can use the loader to load these models and reparent them into the scene.

The rear axle and the wheel centre are both at (0,0,0) in each case. The distance between the axles is about 130 units without scaling. Write a python panda3d program that will draw a bicycle by creating a "bicycle" node and then attaching the frame and two wheels as child nodes to it. You should be able to make a scene similar to the image on the right.

Exercise 2 Color



You should be able to set color attributes for the nodes to make the bicycle look less washed-out as on the right

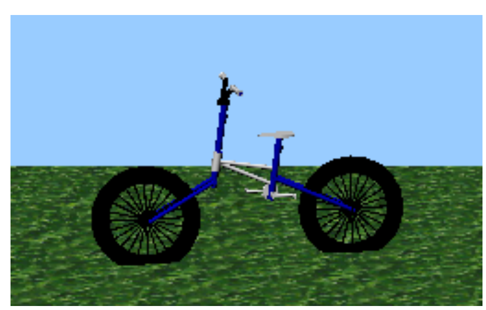
Exercise 3 Intervals

Use an hprInterval to make the wheels of the bike rotate. You can look at the sample/solar-system for some guidance on this or similar demos. You may need to experiment to get it to look right.

Exercise 4 Make the bike move in a circle

If you reparent the bicycle node to another node (e.g. "circle centre"), you can make the bicycle move around it by applying an hprInterval to the circle centre node. The bicycle node will need an offset (setPos) from the parent centre location in order for an hprInterval to make it move around. You make want to look at the sample/solar system for guidance on this too. You may also need to re-orient the bicycle to make it look right.

Exercise 5 Make it look good however you like



The models actually include texture data that will show up if you apply a light source. You could also consider adding a ground surface to your scene. You might also consider repositioning the camera or having it follow the bicycle. Another consideration for realism is that the rate of rotation of the wheels should be such that the bike tires move (velocity *not* rotation) at exactly the same rate as the bike (i.e. zero motion relative to the ground). You can include comments in your python regarding how you did matching like this if you like.